

Лабораторная работа № 5

Эмуляция и измерение задержек в глобальных сетях

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Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

Выполнение лабораторной работы

```
root@mininet-vx:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 2a:0a:82:b0:8d:e1 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1366 bytes 170400 (170.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1366 bytes 170400 (170.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vx:/home/mininet#
```

```
root@mininet-vx:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether c6:ed:47:60:d8:70 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1276 bytes 162328 (162.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1276 bytes 162328 (162.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vx:/home/mininet#
```

Выполнение лабораторной работы

```

"host: h2"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1366 bytes 170400 (170.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1366 bytes 170400 (170.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=4.17 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.217 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.034 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.050 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.052 ms

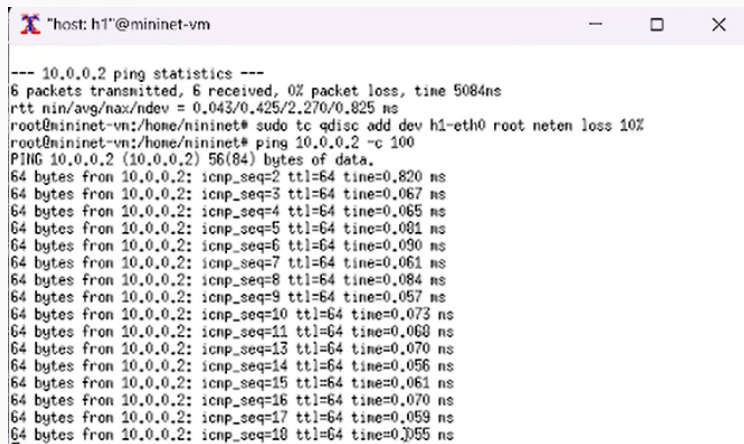
--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5087ms
rtt min/avg/max/mdev = 0.034/0.767/4.172/1.523 ms
root@mininet-vm:/home/mininet#
MAGIC-COOKIE-1 9619d1d4c7047ef5960e725dac8319eb

"host: h1"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1276 bytes 162328 (162.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1276 bytes 162328 (162.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.27 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.043 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5084ms
rtt min/avg/max/mdev = 0.043/0.425/2.270/0.826 ms
```

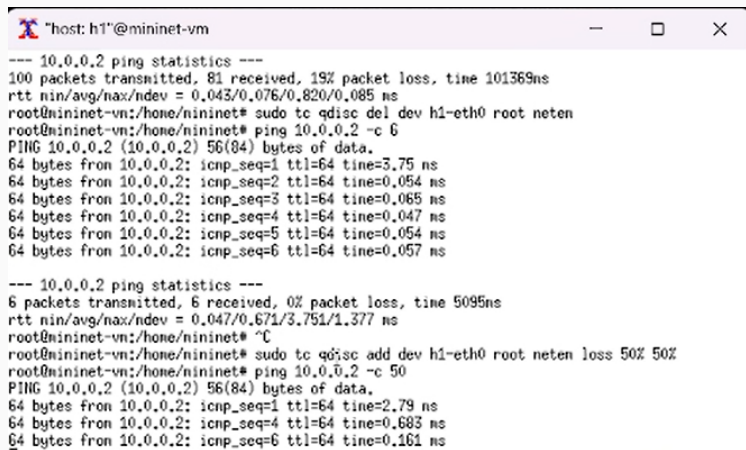


```
host: h1" @mininet-vm
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5084ns
rtt min/avg/max/ndev = 0.043/0.425/2.270/0.825 ns
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.820 ns
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.067 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.065 ns
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.081 ns
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.090 ns
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.061 ns
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.084 ns
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.057 ns
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.073 ns
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.068 ns
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.070 ns
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.056 ns
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.061 ns
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.070 ns
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.059 ns
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.055 ns
```

Рис. 3: Добавление потери пакетов

Выполнение лабораторной работы

```
16 "host: h2"@mininet-vm
TU rtt min/avg/max/ndev = 0,034/0,767/4,172/1,523 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem loss 10%
min: root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 100
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0,922 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0,433 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0,239 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0,082 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0,072 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=0,066 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=0,065 ms
64 bytes from 10.0.0.1: icmp_seq=9 ttl=64 time=0,054 ms
64 bytes from 10.0.0.1: icmp_seq=12 ttl=64 time=0,057 ms
64 bytes from 10.0.0.1: icmp_seq=13 ttl=64 time=0,065 ms
64 bytes from 10.0.0.1: icmp_seq=14 ttl=64 time=0,071 ms
64 bytes from 10.0.0.1: icmp_seq=15 ttl=64 time=0,068 ms
64 bytes from 10.0.0.1: icmp_seq=17 ttl=64 time=0,060 ms
64 bytes from 10.0.0.1: icmp_seq=18 ttl=64 time=0,069 ms
64 bytes from 10.0.0.1: icmp_seq=19 ttl=64 time=0,065 ms
64 bytes from 10.0.0.1: icmp_seq=21 ttl=64 time=0,057 ms
64 bytes from 10.0.0.1: icmp_seq=22 ttl=64 time=0,059 ms
64 bytes from 10.0.0.1: icmp_seq=23 ttl=64 time=0,049 ms
64 bytes from 10.0.0.1: icmp_seq=24 ttl=64 time=0,138 ms
:11 MIT-MAGIC-COOKIE-1 9619d1d4c7047ef5960e725dac8319eb
"host: h1"@mininet-vm
-- 10.0.0.2 ping statistics ---
packets transmitted, 6 received, 0% packet loss, time 5084ns
tt min/avg/max/ndev = 0,043/0,425/2,270/0,825 ms
oot@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10%
oot@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
ING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
4 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0,820 ms
4 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0,067 ms
4 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0,065 ms
4 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0,081 ms
4 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0,090 ms
4 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0,061 ms
4 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0,084 ms
4 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0,057 ms
4 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0,073 ns
4 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0,068 ns
4 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0,070 ns
4 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0,056 ns
4 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0,061 ns
4 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0,070 ns
4 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0,059 ns
4 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0,055 ns
```



```
"host: h1"@mininet-vm
--- 10.0.0.2 ping statistics ---
100 packets transmitted, 81 received, 19% packet loss, time 101369ns
rtt min/avg/max/ndev = 0.043/0.076/0.820/0.085 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.75 ns
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.054 ns
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.065 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.047 ns
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.054 ns
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.057 ns

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5095ns
rtt min/avg/max/ndev = 0.047/0.671/3.751/1.377 ms
root@mininet-vm:/home/mininet# ^C
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 50% 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 50
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.79 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.683 ns
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.161 ns
```

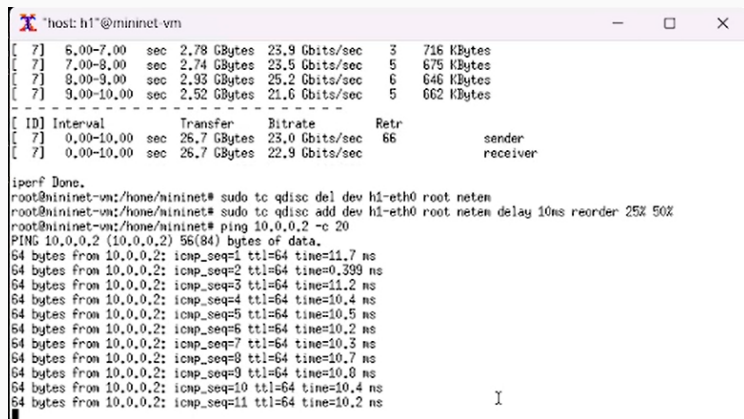
Рис. 5: Добавление значения корреляции для потери пакетов

Выполнение лабораторной работы

```
host: h2" @mininet-vm
Server listening on TCP port 5001
TCP window size: 85.3 KByte (default)
-----
root@mininet-vm:/home/mininet#
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 59620
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 59630
[ ID] Interval           Transfer             Bitrate
[ 7] 0.00-1.00 sec      2.01 GBytes         17.2 Gbits/sec
[ 7] 1.00-2.00 sec      2.46 GBytes         21.2 Gbits/sec
[ 7] 2.00-3.00 sec      2.81 GBytes         24.0 Gbits/sec
[ 7] 3.00-4.00 sec      2.85 GBytes         24.6 Gbits/sec
[ 7] 4.00-5.00 sec      2.85 GBytes         24.5 Gbits/sec
[ 7] 5.00-6.00 sec      2.75 GBytes         23.6 Gbits/sec
[ 7] 6.00-7.00 sec      2.78 GBytes         23.9 Gbits/sec
[ 7] 7.00-8.00 sec      2.74 GBytes         23.6 Gbits/sec
[ 7] 8.00-9.00 sec      2.93 GBytes         25.2 Gbits/sec

last login: T
mininet@mininet-vm$
mininet-vm$ sudo iperf3 -c 10.0.0.2
54 bytes from 10.0.0.2: icmp_seq=43 ttl=64 time=0.089 ns
54 bytes from 10.0.0.2: icmp_seq=44 ttl=64 time=0.085 ns
54 bytes from 10.0.0.2: icmp_seq=45 ttl=64 time=0.133 ns
54 bytes from 10.0.0.2: icmp_seq=46 ttl=64 time=0.056 ns
54 bytes from 10.0.0.2: icmp_seq=47 ttl=64 time=0.046 ns
54 bytes from 10.0.0.2: icmp_seq=48 ttl=64 time=0.109 ns
mininet@mininet-vm$
mininet-vm$ sudo ping -c 50 10.0.0.2
--- 10.0.0.2 ping statistics ---
50 packets transmitted, 27 received, 46% packet loss, time 50134ms
rtt min/avg/max/ndev = 0.045/0.195/2.786/0.521 ms
mininet@mininet-vm$
mininet-vm$ sudo tc qdisc del dev h1-eth0 root netem
mininet-vm$ sudo tc qdisc add dev h1-eth0 root netem corrupt 0.01%
mininet-vm$ sudo iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 59630 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer             Bitrate      Retr      Cwnd
[ 7] 0.00-1.00 sec      2.02 GBytes         17.3 Gbits/sec    12      1.51 MBytes
[ 7] 1.00-2.00 sec      2.47 GBytes         21.2 Gbits/sec     3      1.45 MBytes
[ 7] 2.00-3.00 sec      2.80 GBytes         24.0 Gbits/sec     8      2.17 MBytes
[ 7] 3.00-4.00 sec      2.86 GBytes         24.5 Gbits/sec     7      2.33 MBytes
[ 7] 4.00-5.00 sec      2.87 GBytes         24.7 Gbits/sec     9      649 KBytes
[ 7] 5.00-6.00 sec      2.75 GBytes         23.6 Gbits/sec     8      604 KBytes
[ 7] 6.00-7.00 sec      2.78 GBytes         23.9 Gbits/sec     3      716 KBytes
[ 7] 7.00-8.00 sec      2.74 GBytes         23.5 Gbits/sec     5      675 KBytes
[ 7] 8.00-9.00 sec      2.93 GBytes         25.2 Gbits/sec     6      646 KBytes
```

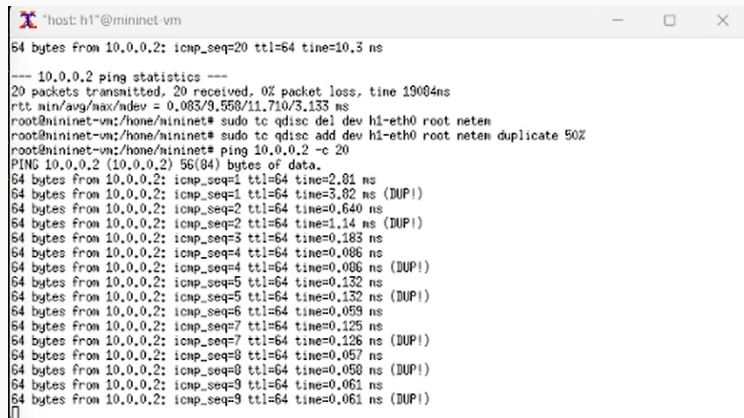
Выполнение лабораторной работы



```
host: h1*@mininet-vm
[ 7] 6.00-7.00 sec 2.78 GBytes 23.9 Gbits/sec 3 716 KBytes
[ 7] 7.00-8.00 sec 2.74 GBytes 23.5 Gbits/sec 5 675 KBytes
[ 7] 8.00-9.00 sec 2.93 GBytes 25.2 Gbits/sec 6 646 KBytes
[ 7] 9.00-10.00 sec 2.52 GBytes 21.6 Gbits/sec 5 662 KBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr
[ 7]  0.00-10.00 sec 26.7 GBytes  23.0 Gbits/sec 66          sender
[ 7]  0.00-10.00 sec 26.7 GBytes  22.9 Gbits/sec          receiver

iperf Done.
root@mininet-vm:/hone/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/hone/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 10ms reorder 25% 50%
root@mininet-vm:/hone/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=11.7 ns
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.399 ns
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=11.2 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.4 ns
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.5 ns
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=10.2 ns
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=10.3 ns
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.7 ns
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.8 ns
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=10.4 ns
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=10.2 ns
```

Рис. 7: Добавление переупорядочивания пакетов



```
*host: h1*@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.3 ns

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19004ns
rtt min/avg/max/ndev = 0.083/9.558/11.710/3.133 ns
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplicate 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.81 ns
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.82 ns (DUP!)
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.640 ns
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.14 ns (DUP!)
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.183 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.086 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.086 ns (DUP!)
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.132 ns
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.132 ns (DUP!)
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.059 ns
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.125 ns
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.126 ns (DUP!)
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.057 ns
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.050 ns (DUP!)
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.061 ns
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.061 ns (DUP!)
^C
```

Рис. 8: Добавление дублирования пакетов

```
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller, waitConnected=True )
    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s1 = net.addSwitch( 's1' )

    info( '*** Creating links\n' )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( '*** Starting network\n' )
    net.start()

    info( '*** Set delay\n' )
    h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem loss 10%' )
    h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem loss 10%' )

    time.sleep(10) # Wait 10 seconds

    info( '*** Ping\n' )
    h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" |
```

```
info( '*** Ping\n')  
h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" |  
awk \'/print $5, $7\\\' | sed -e \'/s/time=//g\' -e
```

Рис. 10: Редактирование скрипта



The screenshot shows a terminal window with the prompt `mininet@mininet-vm: ~/work/lab_netem_ii/simple-drop`. The GNU nano 4.8 editor is open, displaying a Makefile. The file content is as follows:

```
all: ping.dat
ping.dat:
    sudo python lab_netem_ii.py
    sudo chown mininet:mininet ping.dat
clean:
    -rm -f *.dat
```

Рис. 11: Makefile

Выполнение лабораторной работы

```
make: *** [Makefile:3: ping.dat] Error 1
mininet@mininet-vx:~/work/lab_nemem_ii/simple-drop$ ^C
mininet@mininet-vx:~/work/lab_nemem_ii/simple-drop$ sudo mn -c
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflow ovs-controller ovs-testcontroller udbwtest mnexec ixs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflow ovs-controller ovs-testcontroller udbwtest mnexec ixs ryu-manager 2> /dev/null
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --if-exists del-br s1
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_.:a-z0-9]+)-eth[[:digit:]]+'
( ip link del s1-eth1; ip link del s1-eth2 ) 2> /dev/null
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel-Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn*
*** Cleanup complete.
mininet@mininet-vx:~/work/lab_nemem_ii/simple-drop$ make
sudo python lab_nemem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Setting packet loss
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 10%,)
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%,)
*** Running ping test
*** h1 : ('ping -c 100 10.0.0.2 | grep "time=" | awk '{print $7, $8}' | sed -e 's/time=//g' > ping.dat',)
```

Рис. 12: Проведение эксперимента

В результате выполнения данной лабораторной работы я получила навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.